

Unified High-Probability Analysis of Stochastic Variance-Reduced Estimation

Zhankun Luo^{1‡}, Antesh Upadhyay^{1‡}, M. Berk Sahin¹, Sang Bin Moon¹,
Anuran Makur¹, Abolfazl Hashemi¹

¹ Purdue University



Problem

Reliable estimation along a moving optimization trajectory. Stochastic optimization must track a **moving population quantity** from noisy samples. Estimator-specific, expectation-based analyses obscure what controls reliability—especially when errors determine whether a constraint is feasible.

Motivation

- Momentum, STORM, PAGE, and SPIDER reuse historical information differently.
- Compare **memory retention**, **estimate refreshes**, and **movement corrections**.

SEPARATE ANALYSES
Estimator-specific guarantees in expectation.

VS. UNIFIED ANALYSIS
Common structure, high-probability control.

Three families, one design space

CORRECTION	RECURSIVE	RESET-BASED
Standard	Momentum	Probabilistic / periodic momentum
Differential	STORM	PAGE / SPIDER
Second-order	SO Momentum	SO PAGE / SO SPIDER

Momentum smooths direct noise but incurs drift bias. **Differential updates** remove this bias using stochastic differences. **Curvature correction** leaves a quadratic residual.

The same framework covers gradients and scalar constraint values, in Euclidean and smooth non-Euclidean geometries.

Method

- Choose predictable memory β_t , reset probability p_t , and correction T_t .
- Reset with a fresh batch, or combine the current oracle with corrected memory.
- Separate conditional bias from centered innovation noise.

$$\begin{aligned} v_t &= G(w_t, \xi_t^B) \quad (b_t = 1) \\ v_t &= G(w_t, \xi_t) + \beta_t \Delta_t + T_t \quad (b_t = 0) \\ \Delta_t &= v_{t-1} - G(w_{t-1}, \xi_t); \quad b_t \sim \text{Bernoulli}(p_t). \end{aligned}$$

Unified error bound • Theorem 4.2

For $e_t = v_t - g(w_t)$, non-reset steps satisfy $e_t = \beta_t e_{t-1} + I_t(T)$. Use bias bound B_t and centered variance proxy Σ_t^2 .

For $C(\delta, \kappa) = \sqrt{\kappa} + \sqrt{3 \ln(1/\delta)}$:

$$\begin{aligned} \|e_t\|_g &\leq C\left(\frac{\delta}{2T}, \kappa\right) \Lambda_{t,r} \frac{\sigma}{\sqrt{B}} \\ &+ \sum_{j=r+1}^t \Lambda_{t,j} B_j + C\left(\frac{\delta}{2T}, \kappa\right) A_t^{-1} \sqrt{V_t}. \end{aligned}$$

- Initialization:** batch error decays with retained memory.
- Bias:** corrections control accumulated predictable drift.
- Fluctuations:** a vector-valued Freedman inequality controls noise over random reset intervals.

Here $r = \tau_{m(t)-1}$, $\Lambda_{t,j} = \prod_{i=j+1}^t \beta_i$, $A_t = \prod_{i=1}^t \beta_i^{-1}$. The bound holds with probability at least $1 - \delta$ when $\sum_{j=r+1}^t A_j^2 \Sigma_j^2 \leq V_t$ under the stated predictability and sub-Gaussian assumptions.

Key Results

FAMILY	MIRROR DESCENT	EXPECTATION CONSTRAINTS
Standard	ϵ^{-4}	ϵ^{-4}
Differential	ϵ^{-3}	ϵ^{-3}
Second-order	ϵ^{-3}	ϵ^{-3}

Dominant oracle rates from Table 3; logarithmic factors suppressed. Each family uses its stated regularity assumptions.

First variance-reduced improvement for expectation constraints

SO WHAT → The same estimator analysis yields high-probability guarantees in both applications.

Headline Numbers

ϵ^{-3}
Constrained oracle complexity
Improves the prior ϵ^{-4} rate; logarithmic factors suppressed.

3 families
3 regimes
9 configurations

Takeaway

Separate memory, resets, and correction to analyze estimator reliability once.

Stronger guarantees for gradient tracking and expectation-constraint estimation.

Reliability requires **predictable parameters**, **sub-Gaussian noise**, and family-specific regularity. In the constrained application, the objective and constraint are convex, the domain is bounded, and stochastic subgradients are bounded and unbiased.